

There is an “I” in “We”: Relatedness Technologies Viewed Through the Lens of the Need for Autonomy

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ABSTRACT

Technologies that emotionally connect over distance have a long tradition in Human-Computer Interaction (HCI). Naturally, the design of such technologies focuses on fulfilling people’s need for relatedness, i.e., the feeling of being in regular, intimate contact with people who care about you. What seems to be neglected is the possibly conflicting need for autonomy, i.e., the feeling of being the cause of one’s own actions instead of feeling externally controlled. In this essay, we first discuss the potential tension between relatedness and autonomy. We then present five previously published relatedness technologies as examples of how neglecting autonomy might lead to unexpected side effects or the absence of positive effects. We suggest that designing for relatedness should be understood as designing for the ongoing negotiation of relatedness (the “we”) and autonomy (the “I”). This negotiation should not be left to people involved alone but should be integral to designing relatedness technologies.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models; Interaction design; Interaction design theory, concepts and paradigms; Human computer interaction (HCI); Interaction paradigms; Collaborative interaction.

KEYWORDS

Relatedness, Autonomy, Relatedness Technologies, Experience Design

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1 INTRODUCTION

In Human-Computer Interaction (HCI), the design of relatedness technologies has a long-standing tradition. Relatedness technologies enable two or more people to interact in ways that provide

them with a strong sense of emotional connectedness and belonging. This helps people to maintain close relationships with others, care for others, and feel cared for even when they are not physically close. Almost 30 years ago, Robert Strong and William Gaver presented three artifacts, Feather, Scent, and Shaker, which were explicitly designed to promote emotional awareness and intimacy over distance [50]. Since then, a multitude of awareness technologies [14, 27, 54, 57] have been proposed and developed, as well as technologies to foster intimate relationships over a distance [12, 18, 20, 52, 58, 59] or to maintain relatedness among distributed families and friends [17, 25, 26, 32, 37]. Recently, the COVID-19 pandemic brought the pivotal role of technology in maintaining private relationships and enabling informal communication and relational practices over distance to the attention of a wider public. It was no longer just couples and families in long-distance relationships or with frequent travelers in their midst who depended on technology to fulfill their need for relatedness, but virtually everyone. All in all, relatedness technologies can be seen as an established and dynamic field of research within HCI with a long tradition and considerable societal relevance [9, 11, 19, 23, 33, 49].

However, it is also true that despite the development of a variety of artifacts, there is still a lack of understanding and theoretical modeling. It seems safe to say that understanding the quality of mediated interaction, in terms of how exactly the experience of relatedness is created by a design, remains a challenge for designers and researchers alike [41]. There are only a few theoretical approaches to the design for relatedness. For example, Van Bel et al. [7] defined *social connectedness* as “a short-term experience of belonging and relatedness” [7:67] to inform the design of technology and to measure its impact [54]. Ijsselstein et al. [24] proposed *connectedness* as a complementary to *social presence* in HCI, and Rettie [43] concluded that *connectedness* is the more fundamental concept than *social presence* or *awareness* because it addresses a basic human need (see [22]). However, beyond this rather piecemeal approach of different more or less well-defined constructs, there is no comprehensive theoretical model. Even worse, most of the previous work makes indiscriminate use of various constructs such as *closeness*, *intimacy*, or *social presence* [13, 16, 41, 56]. Similarly, *connectedness* is used in a variety of ways, often spontaneously and without reference to psychological or other social science knowledge [10, 37, 40].

To date, only a few literature reviews have taken a systematic look at the field. Hassenzahl et al. [23] described strategies designers use to create the experience of relatedness, such as enabling *emotional expression* or *joint action* through technology, and their underlying psychological principles. Stepanova et al. [49] collected theories from psychology, cognitive science, anthropology, and

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linguistics relevant to the design of relatedness technologies as a theoretical repository. Other reviews provide an overview of design attributes applied [33], derive design considerations [9], principles [11] or factors [19] to provide designers with a more structured approach to the design of relatedness technologies.

In summary, the many existing examples of potential relatedness technologies and the few conceptual approaches remain largely unconnected. In the absence of any overarching model, patterns and influencing factors remain unclear. This runs the risk of overlooking crucial aspects in the design of relatedness technologies. The aim of this essay is to focus on the need for autonomy as one such often overlooked aspect, and to discuss its potential role in the design of successful relatedness technologies.

We first explore the role of autonomy for relatedness. We then present five previously published relatedness technologies as examples of how neglecting autonomy may lead to unexpected side effects or the absence of positive effects. Finally, we discuss how this can inspire future work that contributes to the design of successful, i.e., psychologically fulfilling, and acceptable, relatedness technologies.

2 THERE IS AN “I” IN “WE”: THE ROLE OF AUTONOMY FOR RELATEDNESS

The experience of relatedness is the “feeling that you have regular intimate contact with people who care about you rather than feeling lonely and uncared for” [48:339]. It is relatively undisputed that it is one of the most important human needs [48]. Defined as “the need to feel belongingness and connectedness with others” [45:73] it is one of the three key needs of *Self-Determination Theory* (SDT) [45]. The experience of relatedness, whether in intimate relationships, family, or with friends, is central to psychological well-being [21, 31, 46]. Technology plays an important role in mediating relatedness experiences. Relatedness is a ubiquitous outcome of the use of interactive technology [21] and an important driver in the choice of communication technology [44].

Of course, people do not thrive on relatedness alone. There are several other needs that humans seek to fulfill through their actions, such as the need for competence, for stimulation, for popularity, for security, for meaning, or for autonomy [15, 21]. The experience of autonomy can be defined as the “feeling like you are the cause of your own actions rather than feeling that external forces or pressures are the cause of your actions” [48:339]. It seems likely that relatedness and autonomy are intertwined, maybe even opposing concepts [39]. Close social ties, as important they are, may easily impose restrictions and be perceived as external pressure, which in turn may lead to heteronomy rather than autonomy.

However, although autonomy is often described as conflicting with relatedness or community, SDT [45] emphasizes that autonomy is not necessarily synonymous with independence or individualism. It conceptualizes autonomy as the desire to act in accordance with one’s integrated self, which does not necessarily imply independence from others. To better understand autonomy in this sense (and very much in line with SDT), the concept of *relational autonomy* from ethics and philosophy [36] provides a valuable perspective. Relational autonomy emphasizes the interdependent

aspects of autonomy. From this perspective, autonomy is developed and exercised in relation to others. In this sense, it is not synonymous with pure individualism. But even if relatedness and autonomy are not to be understood as antagonistic needs, an application to the design of relatedness technology suggests that they are at least interrelated, and indeed there is evidence of their interplay in relationship maintenance practices [29]. When using technology to experience relatedness to others, autonomy would imply that both partners experience their actions as self-determined and not controlled by the technology or by the demands of the other. This notion of relational autonomy is rarely addressed in the context of HCI [28]. As mentioned before, autonomy is one of the three basic needs of self-determination theory, and although SDT is frequently used in HCI research [6], autonomy rarely seems to be at the focus of interest. SDT has been widely and increasingly applied to games [51], wellbeing [42], and many other domains [6], but only rarely to relatedness technologies. In general, the application of SDT within HCI has been rather superficial [6], where SDT’s three human needs seem to be used primarily as overarching guides when the goal is to foster or preserve intrinsic motivation [35, 53].

Overall, most of HCI research on relatedness technology does not seem to have considered the need for autonomy in depth. If at all, it is discussed here and there [26, 54], sometimes more in the context of privacy [34, 38], and very rarely explicitly addressed to inform design [9]. *Clique Trip*, an in-car system designed to create a sense of relatedness among friends in different cars [30], deliberately addressed the tension between feelings of separation and closeness. It establishes an audio communication between two different cars only when the cars are physically close to each other. When connected, people in one car can hear everything that is happening in the other car, and vice versa. However, by changing the distance between the two cars, the drivers can subtly control whether everyone is together or in separate cars. In this example, the negotiation of autonomy versus relatedness is an integral part of the design. This is necessary because explicitly expressing the need for autonomy in a social situation can be seen as problematic. Imagine that audio communication is realized by simply turning it on or off. Let’s say passengers in car A start annoying passengers in car B. Someone in car B would then have to deliberately disconnect, which would have to be explained or even justified to the others. In the present design, distance provides a subtle mechanism for either intensifying relatedness or taking an autonomy-refreshing break. Another example is a sensory-based awareness system for seriously ill patients and their families [55]. An artificial flower glows to indicate the person’s activity, and by touching the leaf, one can express a desire to talk using conventional devices. Knowing that these patients need to take a break, the system subtly helps them negotiate relatedness and autonomy – all they have to do is take the petal off the flower.

In summary, the need for autonomy is rarely considered in the design of relatedness technology. This is particularly surprising, given its central role in SDT in its presumed link to relatedness. In the following section, we take a closer look at the empirical findings of five previously published relatedness technologies to explore whether the absence of positive effects or unexpected side effects could be explained by a conflict with the need for autonomy.

3 FIVE EXAMPLES

In this section, we present five examples to illustrate the importance of considering autonomy in the design of relatedness technologies. We examine the empirical results of five different prototypes that have been tested “in the wild” through the lens of the need for autonomy. We came across these examples while preparing a systematic review of relatedness technologies. They caught our attention, as we tried to better understand the results. In the following, we attempt to provide a plausible explanation for why the technologies in these examples had surprising side effects or did not convey only positive experiences. Specifically, we highlight five different themes that merit further discussion: “Reflection of the Other by Limiting Autonomy,” “Shared Experience as a Function of Synchronous Behavior,” “Feeling Controlled by the Other or Technology,” “Desire for Intimacy While Respecting Individual Autonomy,” “Enabling Autonomy of Interaction”.

3.1 Reflection of the Other by Limiting Autonomy

The technology probe *SonicAIR* [4] connects people acoustically by transmitting symbolic or generic sounds for selected kitchen activities. For example, opening the refrigerator or pantry door triggers a corresponding sound in the remote participant’s home and vice versa. This *awareness* of domestic activities is meant to support the sociability of independently living older adults. A four-week field study of *SonicAIR* with two befriended older adults and two family members found that the participants were worried about disturbing each other. This led them to refrain from performing kitchen activities at night, such as opening the refrigerator or heating milk in the microwave. They also developed workarounds during the day to accommodate each other’s routines, such as not storing food in the pantry that might be needed in a short time to avoid opening the pantry door repeatedly or collecting and returning all items from and to the refrigerator at once, rather than taking bits and pieces as needed.

While this may not necessarily be detrimental to feeling more related and in-sync with each other, it still hints at restrictions of personal autonomy that seem worth considering. One could even say that this example illustrates very well the difference between independence, relatedness, and relational autonomy. Whereas before the two felt completely independent in their domestic rhythms, the acoustic connection led them to reflect on their own behavior, as one usually does automatically when living with others, without this necessarily leading to unpleasant feelings. In addition, the nighttime kitchen sounds generated new experiences of relatedness. For example, they were interpreted to mean that the other person could not sleep either and the participant allowed herself to get up and watch television. In this way, thinking in terms of the other person’s experience can influence behavior far beyond the interaction with the technology. The authors of *SonicAir* leave it open whether restricting one’s behavior so as not to disturb the other is merely an unwanted side effect or perhaps an integral part of the way *SonicAir* creates relatedness. If the latter were the case, it could be argued that, from a psychological perspective, *awareness* in general uses rather symbolic restrictions of autonomy (of the “I”) to instill deeper reflections about similarities and differences

between the co-inhabiting individuals as a way of creating the experience of relatedness (of the “We”).

3.2 Shared Experience as a Function of Synchronous Behavior

Somnia is a prototype [47] consisting of two internet-connected pillows. When one partner goes to bed, the heating pad in the other pillow heats up, pre-warming the partner’s bed and serving as a social trigger to encourage the partner to go to bed as well. This *awareness* system is designed to help romantic couples who are separated from each other to fall asleep more quickly and to improve sleep quality. During the two-week test with two romantic couples using a working prototype, the participants reported a sense of presence when they noticed the warm bed as a signal that their partner was about to go to sleep. However, no convergence of sleep times was observed. In contrast, the prototype turned out to be a source of frustration. The two female participants did not enjoy the pre-warmed bed on most nights, because they were always the first to go to bed. They reported disappointment at not being able to share this moment with their partner. The authors’ hypothesis that the use of *Somnia* would automatically synchronize bedtimes was not confirmed, although it seemed to be supported by a previous quantitative online survey. In this survey, the authors showed a video prototype presentation to 76 potential users and found that more than 75% of respondents said they would go to bed earlier if they knew their partner’s bedtime. However, the interviews with the couples revealed that external circumstances, such as work activities, interfered with adjusting to their partner’s bedtime.

From an autonomy perspective, the more unbalanced – or unsynchronized – bedtime behavior of couples when separated is quite plausible since the absence of a partner provides room for autonomy. Unfortunately, this resulted in two completely different individual experiences when using the prototype. What was a positive “warm” experience (of the “we”) for one partner, such as cuddling up in the warm bed of the loved one, was a negative “cold” experience of being alone (and passively waiting without being able to tell the partner to come, as one of the female participants described it) for the other, who was disappointed to forgo the “we”. While going to bed early was probably even worse than without the *Somnia* system, being late was rewarded with a warm bed. One can speculate, that the partner’s failure to arrive on time may also have triggered uncomfortable thoughts about why the partner is late. Reflecting on the other’s “autonomous” behavior may not only have led to feelings of detachment but may also have taken on unwanted meanings.

The *Somnia* authors do not discuss the discrepancy between the online survey respondents’ intentions to adjust their own behavior towards shared bedtime practices and the participants’ actual failure to synchronize individual bedtimes and the resulting frustration (at least for one member of the couple). While participants dutifully reported external difficulties as a barrier to synchronizing bedtimes, it seems quite plausible that one partner enjoyed the temporary autonomy. From a psychological perspective, *awareness* systems that rely on synchronization may be less appropriate since temporary separation in close couple relationship also provides a valuable opportunity to live out one’s autonomy. This, in turn is a source of

uncomfortable thoughts for the other partner, which can result in feelings of alienation rather than closeness. In the case of *Somnia*, the authors instrumentalize the relationship rather than designing for it. Instead of focusing on the intricacies of couples, they want to use the relationship to establish better sleep. This could have been done much more easily, and perhaps more successfully, without connecting the devices at all. For example, if the system were to activate itself around a couple's usual bedtime, this simulation of the partner (e.g., warmth) could become a motivation to stick to established practices even when the partner is not there, creating a sense of relatedness independently of what the partner is doing.

3.3 Feeling Controlled by the Other or Technology

The *InTouch* system [3] is an application that runs on an older adult's tablet (and on a family member's laptop or mobile phone) and integrates key communication functions such as sending and receiving photos, videos, and audio messages, with a focus on accessibility. As a special feature, older adults can send "waves", a predefined message to let others know they are thinking of them. This *emotional expression* is designed to better integrate old family members into the communication activities of the rest of the family. In a qualitative pilot study, an older adult and her daughter tested the prototype for seven weeks. While the mother had an overall positive experience, the daughter developed negative emotions such as anger and annoyance rather than emotional closeness. She felt overwhelmed by her mother's demands, because her mother sent her "waves" so many times a day, and the system only offered her the option of responding via her email client. This seemed a rather cumbersome way to respond to her mother's symbolic "I'm thinking of you" message.

From a psychological perspective, the focus of this *emotional expression* system was one-sided, entirely focused on the involvement of the older adult and the accessibility of the technology. The role of the mother's counterpart, the daughter, was approached in a purely functional way. This reproduces the certainly inappropriate notion of an asymmetric caregiving relationship between the two, while, especially in informal care, the emotional relationship between parents and caregiving children is much more complex and two-sided [1]. In other words, the persons' specific needs for relatedness and autonomy were not considered. The daughter's negative experiences are the consequence of feeling controlled by her mother and the technology. This lack of individual autonomy may have prevented the emergence of a sense of emotional closeness and relatedness on her part.

A similar phenomenon, but with less negative feelings, was also apparent in the evaluation of the *Messaging Kettle* [2], a prototype for connecting distributed family members through technologically augmented kettles that indicate use to the remote partner through an animated glow. An additional device for exchanging short handwritten messages, the *Smart Tea Box*, is included in the system. During the long-term in-home study with four pairs of users (mothers and their adult daughters), involving a total of 20 family members, one of the mothers in the test couples was sometimes overwhelmed by the message exchange, because her daughter was sending so many messages and she did not know what to write.

Both examples show that sudden "outbursts of affection", presumably arising from a strong momentary need for relatedness, can leave the target person feeling overwhelmed. Consequently, where there is love and affection on one side, the corresponding technologically mediated behavior and interaction can lead to a negative experience of excessive demands on the other side, violating the other person's sense of autonomy. Interacting then feels obligated, pressured, and filled with negative feelings, rather than as an act of self-determination.

The authors of *InTouch* shared deep insights from previous research on intergenerational differences in communication needs and patterns, and concluded that, among other things, the experience of participating family members needs to be improved, but they do not mention autonomy. From a psychological perspective, a fulfilling experience of relatedness for all connected users would not be a matter of deciding whether or not to apply well-known principles such as *reciprocity* and *asynchronicity*, but of supporting users to find and negotiate the appropriate level of relatedness and autonomy that meets their individual, momentary needs. This would imply that a "well-designed" *emotional expression* system should not only trigger interaction, but also suggest appropriate forms of interaction to keep interaction in proportion. This protects individual autonomy without sacrificing relatedness.

3.4 Desire for Intimacy Without Respecting Individual Autonomy

The technology probe *In your eyes* [5] connects couples through always-on video and audio streaming. The prototype consists of a smartphone placed in a shirt pocket with the camera facing out. It runs video chat software in auto-response mode. One partner is able to "visit" and participate in the daily activities of the other at any time and can also actively speak up and join the situation at any point, without explicit permission. The designers' intention of this *awareness and joint action* device was to help long-distance couples stay deeply connected over distance and share their daily experiences to support relationship maintenance. It was evaluated in a one-month qualitative field study with two couples, both in a long-distance relationship. They were both confronted with challenging situations; in particular, spontaneous, unpredictable visits were not well received by those visited. The couple who actively talked about their negative experiences to adjust future behavior, such as not calling in official situations (although the girlfriend seemed to take some pleasure in poking her partner in such situations), was able to cope and benefit, and the technology probe contributed to relatedness. The other couple felt intrusively over-connected and disconnected at the same time, with negative effects on their sense of relatedness and even their relationship.

As the authors pointed out, a plausible reason for the negative experience of (over)connectedness may have been that the called person felt out of control and that the partner's behavior violated individual autonomy. While from a romantic perspective this system literally follows the idea of being inseparable, it completely neglects (as new lovers may foolishly do) each other's need for autonomy. As a consequence, there is "we" all day long, even in situations, where the active partner needs to focus on themselves or

others. Especially in close relationships, the balance between relatedness and autonomy has to be constantly negotiated [29], and not every couple prefers to do so explicitly. The design of *In your eyes* did not provide a built-in way to empathetically approach the partner, nor did it provide a way to respectfully exclude the other person in situations that a partner did not want to share. In addition, the design of “sitting in your sweetheart’s breast pocket” (instead of, for example, obviously carrying the other person around with a telepresence robot) further dissolves the boundaries between the two people. The partners literally become one, which may be the stuff of romantic movies, but is certainly not beneficial to a healthy, long-lasting relationship [8].

One can speculate whether this extreme use of conventional technology, as the authors call it, would have been designed differently if the need for autonomy had been considered as important as relatedness. In fact, the radical nature of this *awareness* and *joint action* device demonstrates that feeling related does not depend solely on the degree of technical connectedness and the amount of time spent together. Even for couples who are deeply in love and have a strong desire for intimacy, the disregard of individual needs for autonomy can have a negative impact on the relationship. The paradox of wanting to reach one’s partner at all times, but not wanting to be constantly connected (and thus available to one’s partner), has also been found in other prototype evaluations with romantic couples [18], and seems to be an inherent challenge for relatedness technologies, that is neither conceptually understood nor adequately supported by the available work.

3.5 Enabling Autonomy of Interaction

Fribo is a prototype [25] that recognizes specific domestic sounds in the user’s home and shares this information with a circle of friends who also use it. A robot embodies the system and converts domestic sounds into spoken messages that are sent anonymously to everyone. Whenever one of the friends performs a pre-defined mundane activity, such as using the washing machine, the robot reports the activity in real time, but without revealing the person who performed it. The role of this *awareness* robot is that of a mediator, telling what is going on in the homes of the group of friends without giving too many details. It was designed to help people who live alone feel more related to their friends and create a sense of shared living (i.e., ambient sociability). Evaluation of *Fribo* with three groups of friends, all living in single-person households, in a four-week in-home field study showed that the system not only promoted feelings of relatedness, but significantly increased other forms of communication. Regarding autonomy needs, the concept seemed to succeed in providing an appropriate level of “intrusion”. None of the participants reported a feeling of being monitored or a violation of their privacy and autonomy. The small but significant fact that the domestic sounds were transmitted anonymously and not directly as “noise” may have prevented the effects reported in the *SonicAir* study. Another factor that may have contributed to the overall positive experience of relatedness was the careful choice of activities, which did not convey information that was too sensitive, but still somewhat newsworthy and capable of generating curiosity.

The authors concluded that the user experience with this *awareness* device was one of curiosity and enjoyment, but do not further

discuss the role of autonomy. In fact, *Fribo* collates all the activities from the friends’ homes into a “one of us” story, rather than a detailed report of what exactly happens in each of the homes, which might have been the key to mitigating conflicts with autonomy. In this way, the announcement of someone else’s activity can lead to a variety of individual reactions (the “I”) including triggers to relate (the “we”): It could be perceived as “nice-to-know,” it could inspire one’s own behavior, or, if one happens to be engaged in the same activity, it could create the feeling of a happy coincidence. Or it could serve as a conversation starter to get in touch with friends directly. From a psychological perspective, the prototype’s success suggests that preserving individual autonomy may be a prerequisite for providing an overall positive experience of relatedness. *Fribo* creates an *awareness* of others that does not require specific or direct interaction, thereby respecting the individual’s sense of autonomy and providing a form of, admittedly weak, but enjoyable, ambient sociability.

4 DISCUSSION

We argue that autonomy is crucial and that it needs to be explicitly addressed in the design of relatedness technologies, regardless of the type of relationship. Ignoring the need for autonomy may be a possible explanation for unexpected and even counterproductive effects. As the prototypes promote and enable new ways and forms of connecting, it seems logical that individual autonomy may be affected. In some situations, the required levels of relatedness and autonomy may not match, such as when one partner has just come home from a busy day of meetings and other social situations, while the other partner has been alone all day. In other situations, one of the partners involved may simply feel uncomfortable (as in official situations) or unable to respond appropriately (because they are busy with something else) to the partner’s interaction. Finally, there is an inherent asymmetry in the interaction that establishes relatedness. While one partner typically initiates an interaction quite autonomously, for example, by sending a “wave,” the other is inevitably controlled by that interaction when it requires a response. All these mismatches require negotiation without running the risk of being interpreted as rejection. All in all, we suggest that the “effectiveness” of a relatedness technology, in terms of an experience that is perceived as positive by all involved and that becomes established as a positive relational practice in everyday life, depends on the extent to which it addresses the need for autonomy of all involved, i.e. how well relatedness and autonomy can be negotiated and/or regulated.

The examples presented provide some evidence that consideration of autonomy would have been beneficial. It also shows that details can make or break a positive or negative experience. Designers and researchers should always keep this in mind when designing relatedness technologies. After all, it is about the most important thing we have as humans: our social relationships. Design can better support this by focusing not only on the obvious goal of increasing social connectedness, but also by respecting and better understanding the role of disconnectedness. Allowing individual autonomy is not a threat to the relationship or to the overall sense of relatedness, it is integral. Designing for relatedness, closeness, or togetherness should therefore be understood as designing for the

ongoing negotiation of relatedness (the “we”) and autonomy (the “I”), and not simply as enabling awareness or communication. This negotiation is difficult. It should not be left to the people involved alone but should become an integral part of the design of relatedness technologies. Otherwise, the technologies run the risk of being unsuccessful or, worse, detrimental to the relationship. The ultimate goal of the design should be for all involved to perceive their actions to establish relatedness as self-determined, i.e., not controlled and externally determined by either the technology or another person. Designing technologies that evoke a sense of self-determined relatedness is a challenge, and examining the impact of autonomy on interpersonal relatedness is therefore a promising avenue for future research.

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